Retention Analysis Summary for SAS PhD Programs

Year-to-Year Retention Rate (Enrolled to Retained Term)

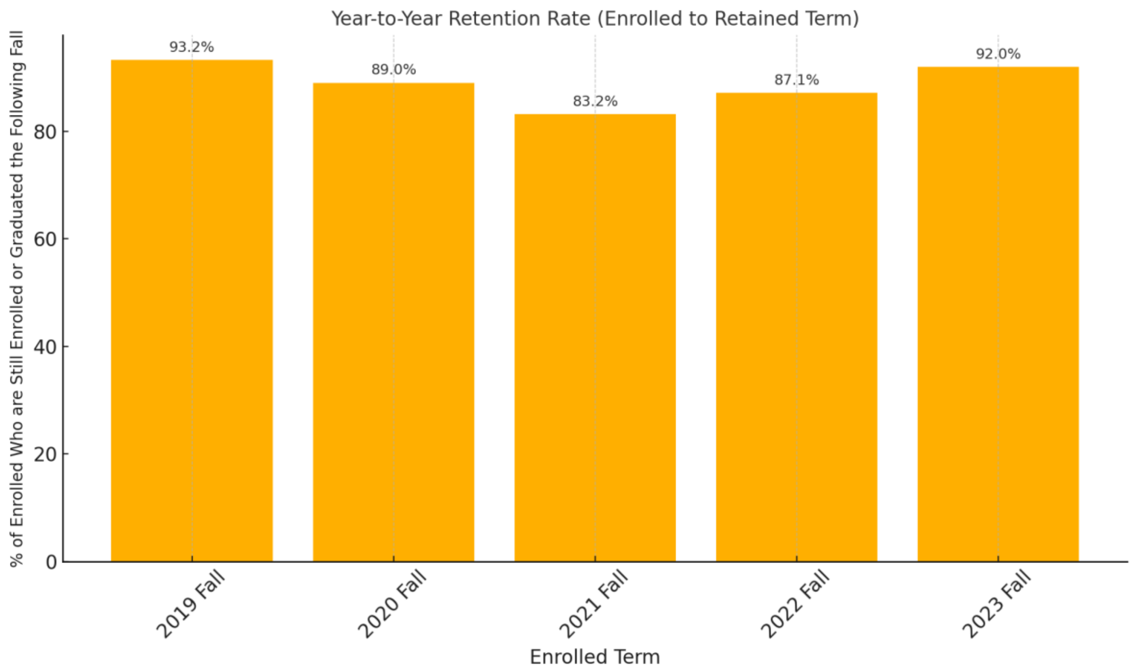
By comparing retention rates from Fall to the following Fall where retention is defined as either remaining enrolled or having graduated, we see a slight uptrend since 2022 Fall Enrolled Term after a downtrend that bottomed 2021 Fall Enrolled Term. However, the only statistically significant difference in retention rates occurred when **comparing the 2019 Fall Enrolled Term and the 2021 Fall Enrolled Term and this represented a decline of 20% in retention.**

Table: Year-to-Year Retention Rates

The table below shows the number of students retained (or graduated) versus not retained for each Enrolled Term.

Enrolled Term	Retained Term	Not Retained	Retained or graduated	Retained or graduated (%)
2019 Fall	2020 Fall	14	193	93.2%
2020 Fall	2021 Fall	24	194	89.0%
2021 Fall	2022 Fall	40	198	83.2%
2022 Fall	2023 Fall	27	183	87.1%
2023 Fall	2024 Fall	17	195	92.0%

Chart: Year-to-Year Retention Rates



This chart shows the percentage of PhD students enrolled in each Fall term who were still enrolled or had graduated (with MA or PhD) the following Fall.

Conclusion

A statistically significant decrease in retention rates (-20.0%) was found between the 2019 Fall Enrolled Term and 202 Fall Enrolled Term for PhD students in SES.

This means it is more likely the actual retention rate decreased as opposed to undergoing normal year-to-year variation in retention.

However, changes in retention rates when comparing other years were not found to be significant. This means the assumption that the retention rate remains the same held true and the more likely explanation of the differences was normal variance in year-to-year enrollment and graduation numbers. As an example, while the retention rate between 2022 Fall to 2023 Fall differed, this difference was not considered significant, and the more likely explanation was an entirely normal year-to-year variation in retention.

Appendix

A Chi-square test of independence (Retained vs Not Retained) was conducted on the contingency table Year-to-Year Retention Rates. A statistically significant change in the retention rate was found at a 0.05 significance level (p-value = 0.0064, df=4). This implied there was at least one pair for years with a significant change in the retention rate.

This led to conducting pairwise retention rate comparisons to see which pairs of years showed a significant difference.

A statistically significant decrease in retention rates (-20.0%) was found from the Enrolled Term 2019 Fall to Enrolled Term 2021 Fall for PhD students in SES.

The table below lists Z-scores and p-values for pairwise comparisons of retention rates between each pair of Enrolled Terms. Adjusted p-values use Bonferroni correction.

Enrolled Term 1	Enrolled Term 2	Z-Statistic	P-Value	P-Value (Bonferroni)
2019 Fall	2020 Fall	1.5333	0.1252	1.0000
2019 Fall	2021 Fall	3.2363	0.0012	0.0121
2019 Fall	2022 Fall	2.0896	0.0367	0.3665
2019 Fall	2023 Fall	0.4909	0.6235	1.0000
2020 Fall	2021 Fall	1.7804	0.0750	0.7502
2020 Fall	2022 Fall	0.5899	0.5552	1.0000
2020 Fall	2023 Fall	-1.0556	0.2912	1.0000
2021 Fall	2022 Fall	-1.1697	0.2421	1.0000
2021 Fall	2023 Fall	-2.7978	0.0051	0.0515
2022 Fall	2023 Fall	-1.6261	0.1039	1.0000